

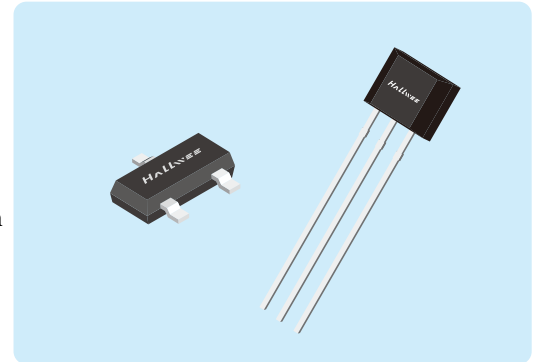
TMR2101 Omnipolar Low-Power Magnetoresistive Switch

1. Overview

The TMR2101 is a magnetic switch sensor that integrates Tunneling Magnetoresistance (TMR) and CMOS technology. It features characteristics such as high precision, high speed, low power consumption, and high sensitivity, making it suitable for magnetic field switch detection in industrial and consumer electronics. The chip's internal circuitry includes a voltage generator, comparator, digital logic control module, threshold trimming module, and CMOS output circuit.

The TMR2101 features a wide operating voltage range and a wide operating temperature range. This series of chips can provide various switching operating frequencies and package types to adapt to various applications.

The TMR2101 is an omnipolar magnetic field detection switch that provides an omnipolar magnetic response with extremely low current consumption. It detects magnetic fields parallel to the chip's package surface; when the magnetic field strength is greater than the operating point (BOP), the switch outputs a low level, and when the magnetic field strength is less than the release point (BRP), the switch outputs a high level. This chip can operate within a supply voltage range of 1.8V to 5.5V and uses standard SOT-23-3L and TO-92S packages.



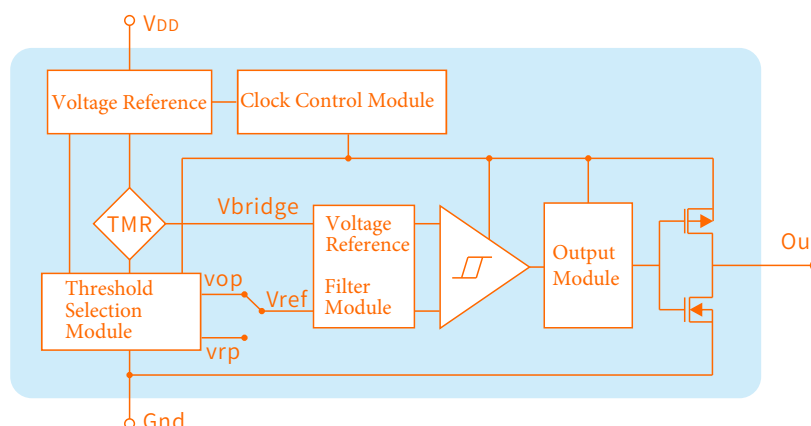
2. Product Features

- ◆ Ultra-low power consumption
 - 50Hz version: 160nA@3.0V
 - 1000Hz version: 3.4uA@3.0V
- ◆ Wide operating voltage range: 1.8V~5.5V
- ◆ CMOS push-pull output
- ◆ Operating temperature range: -40°C~125°C
- ◆ Excellent ESD performance: HBM 8KV

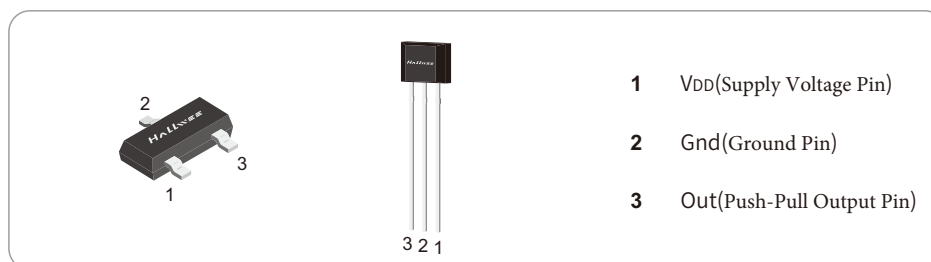
3. Typical Applications

- ◆ Water meters, gas meters, flow meters
- ◆ Non-contact detection
- ◆ Electronic locks, valve position detection
- ◆ Laptop and tablet switch detection
- ◆ TWS earphones, mobile phones

4. Functional Block Diagram



5. Pin Definitions



6. Absolute Maximum Ratings

Parameter	Symbol	Value	Unit
Supply Voltage	V_{DD}	6	V
Reverse Supply Voltage	V_{DD_REV}	-0.3	V
Output Drive Current	I_{OUTPUT}	5	mA
Maximum Magnetic Field Exposure	B	3000@<5min	Gs
Package Power Dissipation	P_D	400	mW
Storage Temperature Range	T_{STG}	-50~+150	°C
Maximum Junction Temperature	T_J	+150	°C
Maximum Reflow Temperature	T reflow	260	°C
ESD Capability (Human Body Model)	ESD HBM	8000	V

Note: Exceeding the absolute maximum ratings may cause permanent damage. Prolonged operation under absolute maximum rating conditions may affect device reliability.

7. Electrical Parameters

TMR2101L 50HZ Version

Parameter	Symbol	Working Condition	Min	Typical	Max	Unit
Supply Voltage	VDD	Active	1.8	—	5.5	V
Output Low Level	VOL	$I_{OUT}=1mA$	—	0.05	0.15	V
Output High Level	VOH	$I_{OUT}=1mA$	$V_{DD}-0.15$	$V_{DD}-0.05$	—	V
Average Current	$I_{DD(AVG)}$	$T_A=+25^{\circ}C$ VDD=3.0V	—	160	—	nA
Awake State Current	$I_{DD(Awake)}$	$T_A=+25^{\circ}C$ VDD=3.0V	—	3.4	—	μA
Sleep State Current	$I_{DD(Sleep)}$	$T_A=+25^{\circ}C$ VDD=3.0V	—	148	—	nA
Awake Time	T_{AWAKE}	Active	—	40	—	μs
Period	T_{PERIOD}	Active	—	20	—	ms

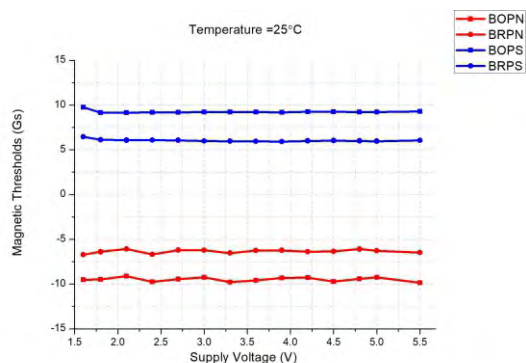
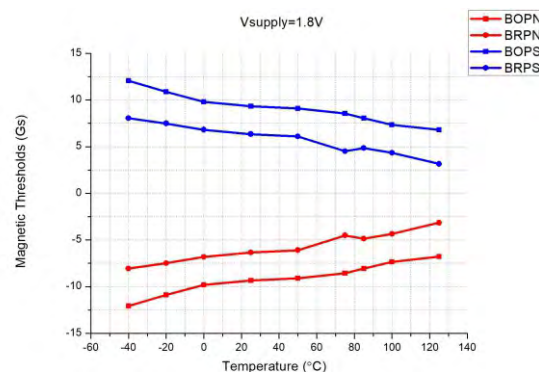
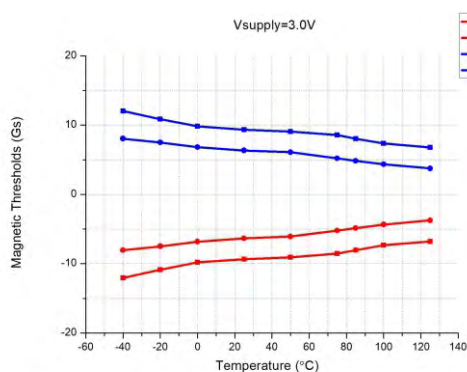
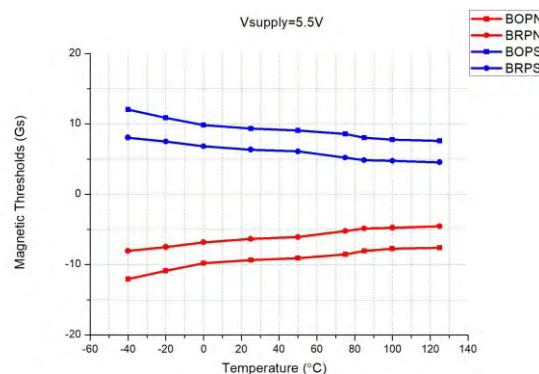
TMR2101H 1000 HZ Version

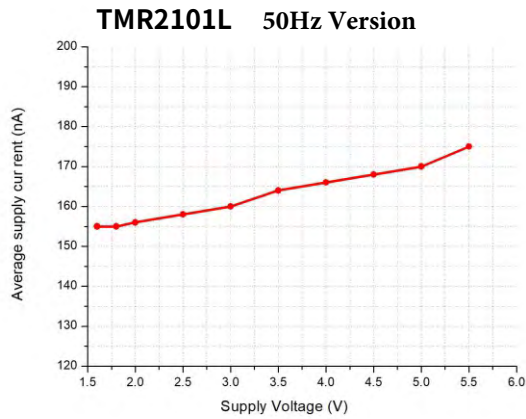
Parameter	Symbol	Working Condition	Min	Typical	Max	Unit
Supply Voltage	VDD	Active	1.8	—	5.5	V
Output Low Level	VOL	$I_{OUT}=1mA$	—	0.02	0.1	V
Output High Level	VOH	$I_{OUT}=1mA$	$V_{DD}-0.1$	$V_{DD}-0.02$	—	V
Average Current	$I_{DD(AVG)}$	$T_A=+25^{\circ}C, V_{DD}=3.0V$	—	3.4	—	μA
Switching Frequency	F_S	Active	—	1000	—	Hz

8. Magnetic Characteristics

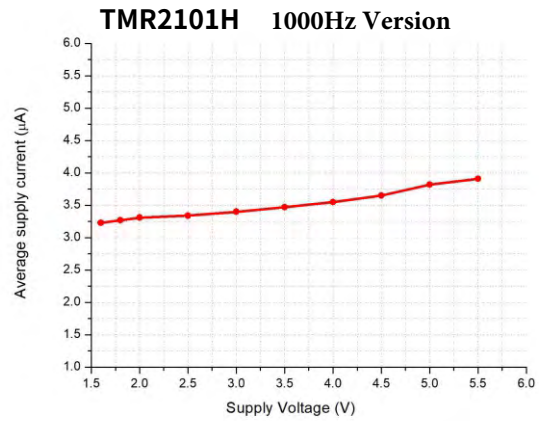
Parameter	Symbol	Working Condition	Min	Typical	Max	Unit
Magnetic Operating Point (South)	B_{OPS}	$T_A=+25^{\circ}C, V_{DD}=3.0V$	6	9	14	Gs
Magnetic Release Point (South)		$T_A=+25^{\circ}C, V_{DD}=3.0V$	3	6	9	Gs
Magnetic Operating Point (North)	B_{OPN}	$T_A=+25^{\circ}C, V_{DD}=3.0V$	-14	-9	-6	Gs
Magnetic Release Point (North)		$T_A=+25^{\circ}C, V_{DD}=3.0V$	-9	-6	-3	Gs
Hysteresis	$B_{HY} (B_{OPX} - B_{RPX})$		-	3	-	Gs

9. Performance Curves


Magnetic Threshold vs. Supply Voltage @ $T_A=25^{\circ}C$

Magnetic Threshold vs. Temperature @ $V_{DD}=1.8V$

Magnetic Threshold vs. Temperature @ $V_{DD}=3.0V$

Magnetic Threshold vs. Temperature @ $V_{DD}=5.5V$

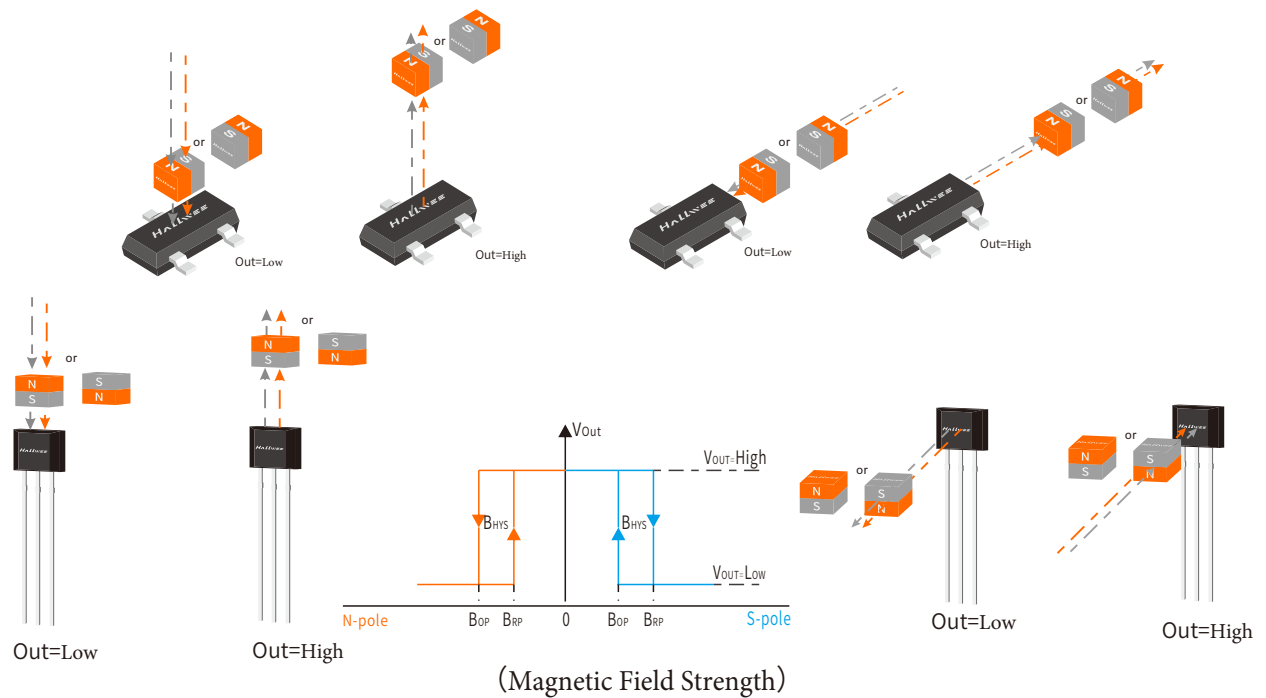


Average Operating Current vs. Supply Voltage @ $T_A=25^{\circ}\text{C}$

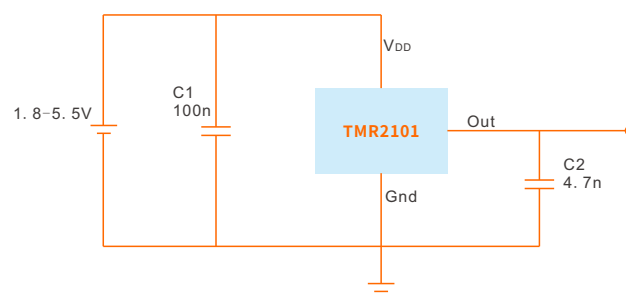


Average Operating Current vs. Supply Voltage

10. Magnetoelectric Conversion Characteristics



11. Typical Application Circuits

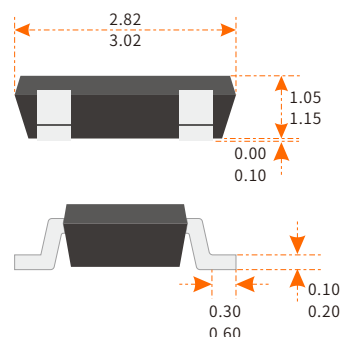
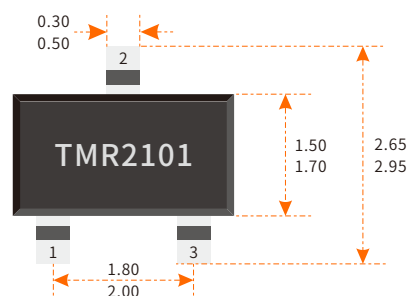


12. Ordering Information

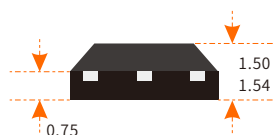
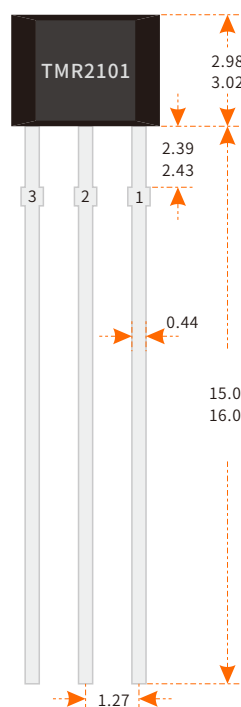
Part Number	Package Type	Minimum Packaging Quantity
TMR2101L SO	SO (SOT-23-3L)	3000PCS
TMR2101L UA	UA (TO-92S)	1000PCS
TMR2101H SO	SO (SOT-23-3L)	3000PCS
TMR2101H UA	UA (TO-92S)	1000PCS

13. Package Dimensions

SOT-23 (Small Outline Transistor) Package



TO-92S (UA Package)



Notes:

1. Measurement Unit: mm
2. Mounting Precaution: Lead pins must stay clear of "Flash" (excess molding material) and electroplating pinholes.
3. Lead Bending: Do not bend the lead wires within 1 mm of the package interface.
4. Pin Assignment:
 - a. Pin 1: Power (VDD)
 - b. Pin 2: Ground (Gnd)
 - c. Pin 3: Output (Out)